

DA 3008 - Big Data Analytics

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Title of the Assignment:

Executive Summary Report: Black Friday Retail Performance Analysis.

This report provides a comprehensive evaluation of consumer shopping behaviors, revenue generation, and tactical sales impacts based on the **retail_black_friday_sales_100k.csv** dataset analyzed within the PySpark environment. The dataset captures 100,000 transactions across diverse geographic regions, customer archetypes, and product portfolios.

1. How Discounts Impact Purchase Decisions

The dataset tracks key metrics including purchase_amount, product_category, age_group, customer_segment, discount_price, and purchase_hour. High value electronics consistently sit at the peak of overall transaction value, suggesting that deep promotional discounting acts as a strong catalyst for unlocking high-ticket impulse buys.

- **High-Ticket Validation:** The top transactions indicate individuals willing to commit to single purchase volumes exceeding \$2,000 when original pricing margins scale up to \$2,499.89.
- **Transactional Drivers:** Discounts mitigate structural friction during promotional spikes like Black Friday, transforming passive browsers into buyers by lowering the final entry barrier on luxury or high-end category inventory.

2. Revenue Generation Across Product Categories

Data processing identifies Electronics as the undisputed anchor of sales performance, yielding more revenue than the next several categories combined.

Category Sales Breakdown

Product Category	Total Revenue Generated (\$)
Electronics	\$14,123,441.47
Home & Kitchen	\$6,461,099.92
Sports	\$3,303,563.29
Accessories	\$2,754,561.54

Footwear	\$2,013,317.39
Clothing	\$1,688,471.88
Toys	\$1,669,209.91
Beauty	\$1,374,751.97
Groceries	\$1,075,986.12
Books	\$661,227.18

3. Customer Segment Responses to Sales

Analyzing sales distribution across demographic variables reveals equal engagement levels, though behavioral variations materialize across structural consumer loyalty tiers.

Demographic Spending Neutrality

- **Gender Groupings:** Total spending shows a uniform spread across genders. **Male** consumers contributed \$11,767,285.96, **Other** accounted for \$11,719,413.02, and **Female** buyers generated \$11,638,931.69. This tight distribution indicates that promotional retail events carry universal appeal across all gender groups.

Response by Customer Segment

- **Returning Customers (\$12.13M):** This segment represents the highest total revenue contribution. They are highly reactive to sales events, using promotions to return to a brand they recognize.
- **New Customers (\$10.57M):** Promotional events serve as a highly effective customer acquisition funnel, bringing in a massive surge of first-time buyers.
- **Loyal Customers (\$8.89M):** Steady and reliable baseline contributors, though outpaced during high-traffic sales by broader customer acquisition segments.
- **VIP Customers (\$3.54M):** The smallest gross contributor group, yet they represent a concentrated pool of high-value individuals with premium average order values.

4. Spending Patterns Across Time and Segments

Temporal Drivers

- **Peak Transaction Hour:** Operational metrics isolate 10:00 AM as the absolute peak time window for transaction volume, registering 4,295 individual transactions. This highlights a strong morning shopping surge as consumers look to secure promotional inventory before items sell out.

Regional Stability

Revenue generation displays exceptional geographic balance across major metropolitan hubs. No single city dominates the market, proving that the underlying logistics and promotional reach are evenly distributed:

- **Atlanta:** \$3,618,348.52
- **Miami:** \$3,566,829.80
- **Dallas:** \$3,547,967.43
- **New York:** \$3,546,500.70
- **Phoenix:** \$3,526,014.43
- **Chicago:** \$3,512,360.63
- **Houston:** \$3,474,957.45
- **Seattle:** \$3,456,842.39
- **San Francisco:** \$3,445,376.30
- **Los Angeles:** \$3,430,433.02

5. Real World Retail Buying Behavior

This PySpark data analysis surfaces several critical conclusions regarding modern omni-channel retail mechanics:

Key Behavioral Takeaway: High volume sales events are defined by category focused spending rather than demographic segmentation. Consumer profiles across various ages, cities, and genders show highly similar buying motivations, focusing heavily on electronics and home essentials.

- **The "Morning Rush" Dynamic:** The heavy concentration of sales at 10:00 AM indicates that time sensitive or limited quantity offers successfully create consumer urgency.
- **Acquisition vs. Retention:** While retaining existing customers remains highly profitable (\$12.13M), the influx of new shoppers (\$10.57M) demonstrates that deep promotional events are essential tools for long-term customer acquisition.
- **Inventory Management Rules:** With Electronics and Home & Kitchen commanding over \$20.5M of total demand, supply chains must prioritize stock depth in these core categories to prevent stockouts and maximize revenue potential.

- End of the Report -